

RAILCAB

Admin Panel & Platform

Complete Requirements Analysis Document

Version 1.0 | March 2026 | CONFIDENTIAL

Prepared for: RailCab (Novelski Technologies)

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1. Executive Summary

RailCab is a multi-sided ride-hailing platform deeply integrated with Indian Railways and IRCTC. The platform connects passengers arriving at/departing from railway stations with cabs (individual drivers and vendor fleets). This document defines the complete requirements for the Admin Panel — a centralised web-based control plane that governs every stakeholder: Administrators, Managers, Accountants, Support Agents, Vendors, and Drivers.

1.1 Scope

- Admin Panel (Web — React/Next.js recommended)
- 10 distinct roles with granular permissions
- Vendor onboarding with fleet management (10-40 vehicles per vendor)
- Individual driver onboarding with multi-step KYC & approval workflow
- AI-assisted dynamic fare engine (traffic, demand, time, availability)
- Full VAPT compliance including OWASP Top-10 controls
- Future-ready microservices architecture

1.2 Stakeholders

Stakeholder	Role	Primary Concern
Super Admin	Full platform control	System stability & compliance
Manager	Operations oversight	Driver & ride management
Accountant	Financial control	Payments, invoices, settlements
Support Agent	Customer service	Dispute resolution
Vendor	Fleet owner (10-40 cabs)	Fleet earnings & driver mgmt
Individual Driver	Self-employed cabbie	Earnings & rides
Passenger (User)	End customer	Booking experience

2. System Architecture Overview

2.1 Component Map

Component	Technology	Purpose
Admin Panel Frontend	React + Next.js + TailwindCSS	Web portal for all admin roles
RailCab Booking Backend	Flask + SocketIO + PostgreSQL + Redis	Core ride engine
IRCTC Proxy Backend	Flask + SQLAlchemy	Train ticket booking integration
User Mobile App	React Native (Expo)	Passenger-facing ride booking
Driver Mobile App	React Native (Expo)	Driver ride management
Fare Engine Service	Python microservice	Dynamic pricing computation
Notification Service	FCM + WebSocket	Real-time alerts
Document Storage	AWS S3 / Cloudflare R2	KYC docs, profile pics
Payment Gateway	Razorpay	Collections & settlements
Cache Layer	Redis	Driver presence, sessions, rate-limits

2.2 Data Flow — Ride Lifecycle

1. User requests ride → RailCab Backend validates → Emits to nearby drivers via Socket.IO
2. Driver accepts → OTP generated → TripHistory created
3. Driver enters OTP → Ride starts → Live location streams to user
4. Driver ends ride → Payment computed → Vendor/Driver settlement queued
5. User & Driver rate each other → Feedback stored
6. Accountant reviews settlements → Approves payouts

3. Role-Based Access Control (RBAC)

3.1 Role Hierarchy

Role	Level	Created By	Max Accounts
Super Admin	1	System / Self	3
Manager	2	Super Admin	Unlimited
Accountant	2	Super Admin	Unlimited
Support Agent	3	Manager / Super Admin	Unlimited
Vendor	4	Self-register (Admin approved)	Unlimited
Vendor Sub-Admin	5	Vendor	5 per vendor
Individual Driver	6	Self-register (Admin approved)	Unlimited
Read-Only Auditor	7	Super Admin	10

3.2 Permission Matrix

Permission	Super Admin	Manager	Accountant	Support	Vendor	Auditor
Create/Delete Roles	☒	☐	☐	☐	☐	☐
Approve Drivers	☒	☒	☐	☐	☐	☐
Approve Vendors	☒	☒	☐	☐	☐	☐
View All Rides	☒	☒	☒	☒	☒ (own)	☒
Manage Fares	☒	☒	☐	☐	☐	☐
Process Payouts	☒	☐	☒	☐	☐	☐
View Financials	☒	☒	☒	☐	☒ (own)	☒
Resolve Disputes	☒	☒	☐	☒	☐	☐
Send Notifications	☒	☒	☐	☒	☒ (own)	☐
Export Reports	☒	☒	☒	☐	☒ (own)	☒
System Config	☒	☐	☐	☐	☐	☐

Permission	Super Admin	Manager	Accountant	Support	Vendor	Auditor
VAPT Audit Logs	☒	☐	☐	☐	☐	☒

3.3 RBAC API

POST /admin/roles — Create Role

Request Body:

```
{
  "role_name": "manager",
  "permissions": ["view_rides", "approve_drivers", "manage_fares"],
  "created_by": "super_admin_id_xxx",
  "description": "Operations manager"
}
```

Response:

```
{
  "role_id": "role_abcl23",
  "role_name": "manager",
  "permissions": ["view_rides", "approve_drivers", "manage_fares"],
  "created_at": "2026-03-01T10:00:00Z",
  "status": "active"
}
```

POST /admin/users — Create Admin User

```
// Request
{
  "name": "Rahul Sharma",
  "email": "rahul@railcab.in",
  "phone": "+919876543210",
  "role_id": "role_abcl23",
  "created_by": "super_admin_id",
  "send_welcome_email": true
}

// Response 201
{
  "admin_user_id": "adm_xyz789",
  "name": "Rahul Sharma",
  "email": "rahul@railcab.in",
  "role": "manager",
  "temp_password_sent": true,
  "must_change_password": true
}
```

4. Module 1 – Super Admin

The Super Admin has unrestricted access to every feature of the platform. Only 1-3 Super Admin accounts should exist. All critical actions must be protected by 2FA and logged immutably.

4.1 Pages & Features

Page	Purpose	Key Actions
Dashboard	Platform-wide KPIs	Live ride count, revenue today, active drivers, pending approvals
User Management	Manage all admin panel users	Create, edit, suspend, delete roles & users
Driver Approvals	Review KYC submissions	Approve / Reject with reason, request re-upload
Vendor Approvals	Review vendor onboarding	Approve / Reject vendor + fleet
Fare Configuration	Set & tune fare algorithms	Base fares, surge rules, geo-zones
Platform Config	System-wide settings	Commission %, OTP expiry, ride radius
Audit Logs	Immutable action history	Filter by user/action/date
Security Dashboard	VAPT & threat monitoring	Failed logins, unusual IPs, rate limit hits
Broadcast	Push notifications	Target users/drivers/vendors by segment

4.2 Dashboard KPIs API

GET /admin/dashboard/summary

Response:

```
{
  "live": {
    "active_drivers": 243,
    "live_rides": 87,
    "pending_requests": 12,
    "drivers_going_online_last_1h": 34
  },
  "today": {
    "total_rides": 1204,
    "revenue": 184600.50,
    "cancellations": 43,
    "avg_fare": 153.32,
    "new_users": 89,
    "new_driver_signups": 7
  }
}
```

```
    },
    "pending_approvals": {
      "drivers": 23,
      "vendors": 4,
      "documents_resubmitted": 6
    },
    "alerts": [
      { "type": "surge_zone", "area": "Hyderabad Central", "multiplier": 1.8 },
      { "type": "low_driver_supply", "area": "Secunderabad", "deficit_pct": 40 }
    ]
  }
}
```

4.3 Platform Config API

GET/PATCH /admin/config

```
// PATCH Request
{
  "platform_commission_pct": 15,
  "vendor_commission_pct": 12,
  "ride_search_radius_km": 5,
  "otp_expiry_seconds": 300,
  "driver_cancel_penalty_inr": 50,
  "user_cancel_penalty_inr": 30,
  "min_driver_rating_threshold": 3.5,
  "max_drivers_per_ride_broadcast": 10,
  "safety_deposit_amount_inr": 2000,
  "auto_suspend_rating_below": 2.5
}

// Response 200
{
  "updated_by": "super_admin_id",
  "updated_at": "2026-03-01T11:00:00Z",
  "config": { /* full config object */ }
}
```


5. Module 2 – Manager

Managers handle day-to-day operational oversight. They approve drivers/vendors, manage fare zones, assign support agents to tickets, and monitor live ride health.

5.1 Pages & Features

Page	Purpose	Input / Actions
Operations Dashboard	Real-time ops view	Live map of drivers, ride heatmap, zone toggles
Driver Management	View all drivers	Filter by city, status, rating; suspend/activate
Driver Approval Queue	KYC review workflow	View docs side-by-side, approve/reject/request info
Vendor Management	Manage vendor accounts	View fleet, performance, commissions
Ride Management	All rides overview	Filter, search, force-complete, cancel, reassign
Fare Zones	Geo-based fare tuning	Draw polygon zones, set surge multipliers
Support Assignment	Ticket management	Assign tickets to support agents, escalate
Shift Management	Driver shift planning	Create shifts, incentive slots
Incentive Programs	Driver bonuses	Create bonus rules (complete N rides → earn bonus)

5.2 Driver Approval Workflow

GET /admin/drivers/pending-approval

```
// Response
{
  "total": 23,
  "drivers": [
    {
      "driver_id": "drv_abc",
      "name": "Mohammed Imran",
      "phone": "+919876543210",
      "vehicle_type": "Cab Economy",
      "vehicle_number": "TS09AB1234",
      "documents": {
        "aadhaar": {"url": "...", "status": "uploaded"},
        "pan": {"url": "...", "status": "uploaded"},
        "driving_licence": {"url": "...", "status": "uploaded"},
        "rc": {"url": "...", "status": "uploaded"},
        "profile_pic": {"url": "...", "status": "uploaded"},
        "vehicle_insurance": {"url": "...", "status": "uploaded"},
        "fitness_certificate": {"url": "...", "status": "uploaded"}
      }
    },
  ],
}
```

```
    "safety_deposit": {"amount": 2000, "status": "completed"},
    "bgv_status": "pending",
    "submitted_at": "2026-03-01T09:00:00Z"
  }
]
```

POST /admin/drivers/:id/approve

```
// Request
{
  "action": "approve",
  "reviewed_by": "manager_id",
  "notes": "All documents verified",
  "bgv_cleared": true
}

// Response 200
{
  "driver_id": "drv_abc",
  "status": "approved",
  "approved_at": "2026-03-01T10:00:00Z",
  "notification_sent": true
}
```

POST /admin/drivers/:id/reject

```
// Request
{
  "action": "reject",
  "reason_code": "DOCUMENT_MISMATCH",
  "reason_detail": "Aadhaar name does not match Driving Licence name",
  "documents_to_reupload": ["aadhaar"],
  "reviewed_by": "manager_id"
}

// Response 200
{
  "driver_id": "drv_abc",
  "status": "rejected",
  "can_reapply_after": "2026-03-08T00:00:00Z",
  "sms_sent": true
}
```

5.3 Ride Management API

GET /admin/rides

```
// Query Params: ?status=pending&city=Hyderabad&from=2026-03-01&to=2026-03-01&page=1&limit=50
// Response
{
```

```
"total": 1204,  
"page": 1,  
"rides": [  
  {  
    "ride_id": "ride_xyz",  
    "user": {"id": "usr_1", "name": "Anitha Reddy", "phone": "+91..."},  
    "driver": {"id": "drv_1", "name": "Imran", "vehicle": "TS09AB1234"},  
    "pickup": "Secunderabad Station",  
    "drop": "Jubilee Hills",  
    "distance_km": 14.2,  
    "fare": 210.40,  
    "status": "started",  
    "vehicle_type": "Cab Economy",  
    "created_at": "2026-03-01T08:45:00Z"  
  }  
]
```

6. Module 3 – Accountant

Accountants manage all financial flows: driver/vendor earnings, platform commissions, GST calculations, payout approvals, and financial reporting. They have READ access to rides but NO ability to modify operational data.

6.1 Pages & Features

Page	Purpose	Key Actions
Financial Dashboard	Revenue overview	Daily/weekly/monthly P&L, commission earned
Driver Earnings	Per-driver earnings ledger	View, export, flag disputes
Vendor Settlements	Per-vendor payouts	Calculate, approve, mark paid
Platform Commissions	Deductions report	Commission per ride, GST breakdowns
Payout Management	Initiate bank transfers	Bulk NEFT/IMPS export, Razorpay payout
Invoice Generator	Auto-generate invoices	Driver/vendor weekly invoices (PDF)
Safety Deposits	Deposit tracking	Collected, refunded, pending
GST Reports	Tax compliance	Monthly GST-R1 export ready data
Dispute Ledger	Financial disputes	Disputed amounts, resolutions, adjustments
Audit Export	CA/auditor exports	Full transaction export (CSV/Excel)

6.2 Earnings & Settlement APIs

GET /admin/earnings/driver/:driver_id

```
// Query: ?from=2026-03-01&to=2026-03-31
// Response
{
  "driver_id": "drv_abc",
  "driver_name": "Mohammed Imran",
  "period": {"from": "2026-03-01", "to": "2026-03-31"},
  "summary": {
    "total_rides": 234,
    "gross_earnings": 38400.00,
    "platform_commission": 5760.00,
    "gst_on_commission": 1036.80,
    "net_payable": 31603.20,
    "already_paid": 20000.00,
    "pending_payout": 11603.20
  },
  "rides": [
    {
      "ride_id": "ride_xyz",
```

```
        "date": "2026-03-01",
        "fare": 210.40,
        "commission": 31.56,
        "net": 178.84,
        "status": "completed"
      }
    ]
  }
```

POST /admin/payouts/initiate

```
// Request
{
  "payout_type": "driver",
  "beneficiary_ids": ["drv_abc", "drv_def"],
  "period_from": "2026-03-01",
  "period_to": "2026-03-07",
  "mode": "razorpay_payout",
  "initiated_by": "accountant_id"
}

// Response 200
{
  "batch_id": "batch_2026_xyz",
  "total_amount": 43206.40,
  "count": 2,
  "status": "processing",
  "estimated_completion": "2026-03-08T18:00:00Z"
}
```

GET /admin/gst-report?month=2026-03

```
// Response
{
  "month": "2026-03",
  "total_taxable_turnover": 2340000.00,
  "output_gst_18pct": 421200.00,
  "input_tax_credit": 12400.00,
  "net_gst_payable": 408800.00,
  "breakdown_by_service": {
    "cab_booking": 350000.00,
    "irctc_convenience_fee": 58800.00
  },
  "gstin": "36AABCT1332L1ZV",
  "export_ready": true
}
```

7. Module 4 – Support

Support Agents handle user and driver complaints, ride disputes, refund requests, and feedback escalations. They have read access to ride details and limited write access to resolve issues.

7.1 Pages & Features

Page	Purpose	Key Actions
Support Dashboard	Ticket overview	Open, pending, resolved counts; SLA breach alerts
Ticket Queue	All support tickets	Filter by type, priority, assignee, SLA
Ticket Detail	Individual ticket view	Conversation thread, ride details, resolve, escalate
Ride Dispute	Dispute resolution	View both sides, adjust fare, initiate refund
User Management	User profile access	View user history, temporary suspend, add note
Driver Issues	Driver complaints	View driver side, flag, send warning
Refund Management	Refund pipeline	Approve/reject refund, trigger Razorpay reversal
FAQ & Scripts	Agent knowledge base	Canned responses, SOPs
SLA Reports	Team performance	MTTR, first response time, resolution rate

7.2 Ticket APIs

POST /admin/support/tickets — Create Ticket

```
// Request
{
  "raised_by_type": "user",
  "raised_by_id": "usr_abc",
  "ride_id": "ride_xyz",
  "issue_type": "overcharged",
  "description": "Driver charged ₹50 more than app showed",
  "priority": "high",
  "attachments": ["https://s3.../screenshot.jpg"]
}

// Response 201
{
  "ticket_id": "TKT-2026-04521",
  "status": "open",
  "assigned_to": "support_agent_id",
  "sla_deadline": "2026-03-01T18:00:00Z",
  "created_at": "2026-03-01T10:00:00Z"
}
```

POST /admin/support/tickets/:id/resolve

```
// Request
{
  "resolution": "Refund approved for overcharged amount",
  "action_taken": "refund",
  "refund_amount": 50.00,
  "resolved_by": "support_agent_id",
  "internal_notes": "Driver had GPS deviation, fare recalculated"
}

// Response 200
{
  "ticket_id": "TKT-2026-04521",
  "status": "resolved",
  "refund_initiated": true,
  "refund_id": "rfnd_abc123",
  "resolved_at": "2026-03-01T11:30:00Z"
}
```

8. Module 5 – Vendor Management

Vendors are fleet operators who register 10 to 40 vehicles under a single business entity. They have their own portal within the admin panel to manage their fleet, drivers, earnings, and payouts.

8.1 Vendor Onboarding Flow

7. Vendor self-registers via vendor portal
8. Uploads business documents (GST, trade licence, PAN, bank details)
9. Admin/Manager reviews and approves
10. Vendor gets access to their fleet management dashboard
11. Vendor adds vehicles and assigns drivers
12. Each driver under vendor still goes through individual KYC
13. Safety deposit per vehicle: configurable in admin

8.2 Vendor Registration Documents

Document	Mandatory	Valid Period
GST Registration Certificate	Yes	Permanent
Business PAN Card	Yes	Permanent
Trade / Shop Licence	Yes	Annual renewal
Bank Account (cancelled cheque)	Yes	Permanent
MSME Registration (if applicable)	No	Permanent
Vehicle Insurance (per vehicle)	Yes	Annual
Vehicle RC (per vehicle)	Yes	As per govt
Driver Licence (per driver)	Yes	As per govt
Fleet Agreement signed with RailCab	Yes	Annual

8.3 Vendor Pages

Page	Purpose	Actions
Vendor Dashboard	Fleet overview KPIs	Total rides today, fleet online %, earnings MTD
Fleet Management	Vehicle list & status	Add/remove vehicle, track live vehicle status
Driver Assignment	Assign drivers to vehicles	Link/unlink driver-vehicle, shift scheduling

Page	Purpose	Actions
Earnings	Revenue breakdown	Per-vehicle, per-driver, period filter
Payouts	Settlement history	View payout history, download invoices
Driver Onboarding	Add driver to fleet	Initiate KYC for new driver under vendor
Documents	Compliance docs	Upload, view expiry alerts
Commission Config	View commission rates	Platform % vs vendor %, dispute commission
Support	Raise vendor issues	Fleet issues, payment disputes

8.4 Vendor APIs

POST /vendor/register — Vendor Self-Registration

```
// Request
{
  "business_name": "Novelski Fleet Pvt Ltd",
  "owner_name": "Suresh Kumar",
  "email": "suresh@novelskileet.in",
  "phone": "+919876543210",
  "gst_number": "36AABCT1332L1ZV",
  "pan_number": "ABCDE1234F",
  "city": "Hyderabad",
  "expected_fleet_size": 20,
  "vehicle_types": ["Cab Economy", "Auto"],
  "bank_account": {
    "account_number": "1234567890",
    "ifsc": "SBIN0001234",
    "account_name": "Novelski Fleet Pvt Ltd"
  }
}

// Response 201
{
  "vendor_id": "vnd_xyz",
  "status": "pending_review",
  "application_number": "VND-2026-0042",
  "estimated_review_days": 2,
  "documents_upload_url": "/vendor/vnd_xyz/documents"
}
```

POST /vendor/:id/vehicles — Add Vehicle to Fleet

```
// Request
{
  "vehicle_number": "TS09AB5678",
  "vehicle_type": "Cab Economy",
  "vehicle_model": "Maruti Swift Dzire",
  "year_of_manufacture": 2022,
  "seating_capacity": 4,
}
```

```
"fuel_type": "CNG",
"rc_document_url": "https://s3.../rc.jpg",
"insurance_url": "https://s3.../insurance.jpg",
"insurance_expiry": "2027-03-01",
"fitness_expiry": "2027-03-01"
}

// Response 201
{
  "vehicle_id": "veh_abc",
  "status": "pending_verification",
  "vendor_id": "vnd_xyz",
  "safety_deposit_required": 2000.00
}
```

GET /vendor/:id/earnings

```
// Response
{
  "vendor_id": "vnd_xyz",
  "fleet_size": 18,
  "active_today": 12,
  "period": "2026-03",
  "gross_earnings": 284000.00,
  "platform_commission_pct": 12,
  "platform_commission_amount": 34080.00,
  "gst_on_commission": 6134.40,
  "net_payout": 243785.60,
  "per_vehicle": [
    {"vehicle_id": "veh_abc", "vehicle_no": "TS09AB5678", "rides": 87, "earnings":
14820.00}
  ],
  "per_driver": [
    {"driver_id": "drv_1", "name": "Ravi", "rides": 45, "earnings": 7650.00}
  ]
}
```

9. Module 6 – Individual Driver Onboarding & Verification

Individual drivers (not under any vendor) go through a stricter multi-step onboarding with human review at each stage, safety deposit collection, and background verification before activation.

9.1 Onboarding Flow (7 Steps)

Step	Action	Owner	System Trigger
1	Driver self-registers (phone OTP)	Driver	SMS OTP + DB entry
2	Personal info submission	Driver	Saved to drivers table
3	Document upload (7 docs)	Driver	Files to S3, status = uploaded
4	Document AI pre-check	System	OCR name match + expiry check
5	Human KYC review	Manager	Approve/reject each doc
6	Safety deposit payment	Driver	Razorpay, status = completed
7	Background verification (BGV)	Manager	Police verification clearance
8	Account activation	System	Status = active, app unlocked

9.2 Required Documents

Document	Verification Method	Auto-Check Fields	Validity
Aadhaar Card (front + back)	OCR + name match	Name, DOB, UID format	Permanent
PAN Card	OCR + name match	Name, PAN format regex	Permanent
Driving Licence	OCR + expiry check	Name, DL number, expiry date	As per validity
Vehicle RC	OCR + owner match	Vehicle no., owner name	As per renewal
Vehicle Insurance	OCR + expiry alert	Policy no., vehicle no., expiry	Annual
Fitness Certificate	Manual review	Expiry date	Annual
Profile Photo	Face clarity check	Blur detection, face detected	Permanent

9.3 Driver Status State Machine

Status	Meaning	Next Possible Status
registered	Phone OTP done, account created	docs_uploaded
docs_uploaded	All 7 docs submitted	under_review / docs_rejected
under_review	KYC in progress	docs_verified / docs_rejected
docs_rejected	One or more docs failed	docs_uploaded (resubmit)
docs_verified	All docs approved	deposit_pending
deposit_pending	Waiting for safety deposit	bgv_in_progress
bgv_in_progress	Background verification underway	active / rejected
active	Fully onboarded	suspended / inactive
suspended	Temporarily suspended	active (after review)
rejected	Permanently rejected	(terminal)

9.4 Driver Onboarding APIs

POST /driver/register

```
// Request
{
  "phone": "+919876543210",
  "otp": "4521",
  "name": "Mohammed Imran",
  "dob": "1990-05-15",
  "city": "Hyderabad",
  "vehicle_type": "Cab Economy",
  "vehicle_number": "TS09AB1234",
  "vehicle_model": "Swift Dzire",
  "referral_code": "REF123"
}

// Response 201
{
  "driver_id": "drv_new123",
  "status": "registered",
  "onboarding_step": 2,
  "next_action": "upload_documents"
}
```

POST /driver/:id/documents — Upload KYC Documents

```
// Multipart form-data OR presigned URL approach
// Request JSON body (after S3 upload)
```

```
{
  "documents": [
    {"type": "aadhaar_front", "url": "https://s3.../aadhaar_front.jpg"},
    {"type": "aadhaar_back", "url": "https://s3.../aadhaar_back.jpg"},
    {"type": "pan", "url": "https://s3.../pan.jpg"},
    {"type": "driving_licence", "url": "https://s3.../dl.jpg"},
    {"type": "rc", "url": "https://s3.../rc.jpg"},
    {"type": "vehicle_insurance", "url": "https://s3.../insurance.jpg"},
    {"type": "fitness_certificate", "url": "https://s3.../fitness.jpg"},
    {"type": "profile_pic", "url": "https://s3.../profile.jpg"}
  ]
}

// Response 200
{
  "driver_id": "drv_new123",
  "status": "docs_uploaded",
  "auto_check_results": {
    "aadhaar_front": {"name_extracted": "MOHAMMED IMRAN", "match": true},
    "driving_licence": {"expiry": "2031-05-15", "valid": true}
  },
  "human_review_eta_hours": 24
}
```

10. Module 7 – Dynamic Fare Engine

The fare engine computes the final ride price using a multi-variable algorithm accounting for distance, time of day, traffic congestion, driver availability, demand level, weather, and special events. All parameters are configurable from the Admin Panel.

10.1 Fare Formula

$$\text{Final Fare} = (\text{Base Fare} + \text{Distance Fare} + \text{Time Fare}) \times \text{Surge Multiplier} \times \text{Demand Factor} + \text{Tolls}$$

10.2 Fare Variables

Variable	Formula / Source	Admin Configurable	Default Value
Base Fare	Fixed per vehicle type	Yes	Bike ₹15, Auto ₹20, Economy ₹40, Premium ₹60, Battery ₹80
Distance Fare	Distance (km) × per_km_rate	Yes	Bike ₹8, Auto ₹10, Economy ₹12, Premium ₹18, Battery ₹25
Time Fare	Minutes in ride × per_min_rate	Yes	₹1.5/min (Economy)
Surge Multiplier	Demand/Supply ratio algorithm	Yes (caps)	1.0 min, 3.0 max
Night Charge	Applied 11 PM – 5 AM	Yes	1.2× multiplier
Weekend/Holiday	Applied on holidays	Yes	1.15× multiplier
Traffic Factor	Real-time congestion API	Yes (weight)	1.0 – 1.5×
Weather Factor	Rain / storm / fog detected	Yes (weight)	1.0 – 1.3×
Booking Ahead	Scheduled rides discount	Yes	0.95× (5% off)
Tolls	Route-based toll data	Partial	Actual toll amount
Platform Fee	Fixed convenience fee	Yes	₹5 per ride

10.3 Surge Algorithm — Supply/Demand Model

14. Every 5 minutes, system counts: active driver sockets in each geo-cell (2km × 2km)
15. Compares with active ride requests + historical demand for same time slot
16. Computes Demand/Supply ratio: D/S
17. Applies sigmoid transformation to get multiplier between 1.0 and max_surge

18. Emits surge zone events via Socket.IO to all nearby driver apps

Surge Multiplier Formula:

```
# Python pseudocode
def compute_surge(demand_count, supply_count, config):
    if supply_count == 0:
        return config["max_surge"]
    ratio = demand_count / supply_count
    # Sigmoid-based smooth surge
    raw_surge = 1.0 + (config["max_surge"] - 1.0) * (1 / (1 + math.exp(-config["k"]
    * (ratio - config["inflection"]))))
    # Clamp to configured bounds
    return round(max(config["min_surge"], min(config["max_surge"], raw_surge)), 2)

# Config defaults
# max_surge = 3.0, min_surge = 1.0, k = 2.0, inflection = 1.5
```

10.4 Admin Fare Configuration Page

Setting	Type	Description
Base Fare per vehicle type	Number (INR)	Starting fare regardless of distance
Per KM Rate per vehicle type	Number (INR)	Distance-based increment
Per Minute Rate	Number (INR)	Time-based increment (traffic)
Max Surge Cap	Multiplier (1.0–5.0)	Hard ceiling on surge
Surge Activation Threshold (D/S ratio)	Float	Min ratio before surge kicks in (e.g. 1.2)
Night Charge Start/End (HH:MM)	Time	When night multiplier applies
Night Multiplier	Float	e.g. 1.2
Rain Surge Enable/Disable	Toggle	Enable weather-based surge
Geo-Zone Overrides	Polygon + overrides	Airport zone, station zone premium fares
Minimum Fare	Number (INR)	No ride below this amount
Fare Review Period	Dropdown	How often auto-recalibration runs

10.5 Fare Calculation API

POST /booking/calculate-fare (Enhanced)

```
// Request
{
  "pickup_lat": 17.4401,
  "pickup_lng": 78.4983,
  "drop_lat": 17.4150,
  "drop_lng": 78.4527,
  "vehicle_type": "Cab Economy",
  "scheduled_time": null,
  "passenger_count": 1
}

// Response 200
{
  "distance_km": 14.2,
  "estimated_duration_min": 28,
  "fares": {
    "Bike": {"base": 15, "distance": 113.6, "time": 28, "surge": 1.0, "total": 156.60},
    "Auto": {"base": 20, "distance": 142, "time": 42, "surge": 1.0, "total": 204.00},
    "Cab Economy": {"base": 40, "distance": 170.4, "time": 42, "surge": 1.2, "total": 303.48},
    "Cab Premium": {"base": 60, "distance": 255.6, "time": 56, "surge": 1.2, "total": 433.92},
    "battery cab": {"base": 80, "distance": 355, "time": 56, "surge": 1.0, "total": 491.00}
  },
  "surge_active": true,
  "surge_reason": "High demand near Secunderabad Station",
  "night_charge": false,
  "toll_estimated": 0,
  "platform_fee": 5
}
```

GET /admin/fare-config — Get Full Fare Config

PATCH /admin/fare-config — Update Fare Config

```
// PATCH Request
{
  "vehicle_rates": {
    "Cab Economy": {"base_fare": 45, "per_km": 13, "per_min": 1.5},
    "Bike": {"base_fare": 15, "per_km": 9, "per_min": 0.5}
  },
  "surge_config": {
    "max_surge": 2.5,
    "activation_threshold": 1.3,
    "k_factor": 2.0
  },
  "night_charge": {"enabled": true, "start": "23:00", "end": "05:00", "multiplier": 1.2},
  "updated_by": "manager_id",
  "effective_from": "2026-03-02T00:00:00Z"
}
```


11. Module 8 – Fleet & Vehicle Management

11.1 Pages

Page	Purpose	Key Actions
Vehicle Registry	All platform vehicles	Filter by type, vendor, city, status
Live Fleet Map	Real-time vehicle positions	Color-coded by status, click for details
Maintenance Alerts	Document expiry reminders	Insurance, fitness cert expiry < 30 days
Vehicle History	Trip history per vehicle	Mileage, earnings, incidents
Deactivate Vehicle	Remove vehicle from platform	Reason, temp/permanent

GET /admin/vehicles — Vehicle List

```
// Response
{
  "total": 342,
  "vehicles": [
    {
      "vehicle_id": "veh_abc",
      "vehicle_number": "TS09AB1234",
      "vehicle_type": "Cab Economy",
      "model": "Swift Dzire",
      "owner_type": "vendor",
      "owner_id": "vnd_xyz",
      "current_driver": "drv_1",
      "status": "on_trip",
      "lat": 17.4401,
      "lng": 78.4983,
      "insurance_expiry": "2027-03-01",
      "fitness_expiry": "2027-01-15",
      "expiry_alert": null
    }
  ]
}
```

12. Module 9 – Analytics & Reporting Dashboard

12.1 Report Types

Report	Metrics Included	Access
Revenue Report	Gross, net, commissions, GST breakdown, period comparison	Admin, Manager, Accountant
Ride Analytics	Total rides, cancellation rate, avg fare, peak hours heatmap	Admin, Manager
Driver Performance	Rides/driver, rating, online hours, income, acceptance rate	Admin, Manager
Vendor Report	Fleet utilisation %, revenue per vehicle, payout history	Admin, Accountant, Vendor (own)
City/Zone Report	Demand, supply, surge events per zone	Admin, Manager
Customer Report	DAU, MAU, retention, repeat ride %, LTV	Admin, Manager
Support Report	Ticket volume, MTTR, top issue categories	Admin, Support Manager
Compliance Report	Doc expiry, KYC completion, BGV status	Admin, Manager

GET /admin/analytics/rides

```
// Query: ?from=2026-03-01&to=2026-03-31&group_by=day&city=Hyderabad
// Response
{
  "period": {"from": "2026-03-01", "to": "2026-03-31"},
  "summary": {
    "total_rides": 38423,
    "completed": 35102,
    "cancelled_by_user": 2100,
    "cancelled_by_driver": 1221,
    "cancellation_rate_pct": 8.65,
    "avg_fare": 182.50,
    "total_revenue": 6401125.00
  },
  "daily": [
    {"date": "2026-03-01", "rides": 1204, "revenue": 184600.50, "cancellations": 43}
  ],
  "peak_hours": [
    {"hour": 8, "rides": 3240},
    {"hour": 18, "rides": 4560}
  ]
}
```

13. Module 10 – Notifications & Communication

13.1 Notification Channels

Channel	Use Case	Admin Configurable
Push Notification (FCM)	Ride updates, offers, alerts	Yes – content, timing, target segment
SMS (Twilio / Fast2SMS)	OTP, ride confirmation, payouts	Yes – templates
In-App Notification	Offers, surge alerts, news	Yes
Email	Invoices, registration, monthly statement	Yes – templates
WhatsApp (optional)	Driver payout reminders	Yes (future)

POST /admin/notifications/broadcast

```
// Request
{
  "title": "Dussehra Special Surge",
  "message": "Expect 1.5x surge tonight 7PM-11PM near Tank Bund. Plan ahead!",
  "target": {
    "audience": "users",
    "city": "Hyderabad",
    "filters": {"last_active_days": 30}
  },
  "channels": ["push", "in_app"],
  "scheduled_at": "2026-10-12T18:00:00Z",
  "created_by": "manager_id"
}

// Response 200
{
  "notification_id": "ntf_abc",
  "target_count": 12430,
  "scheduled_at": "2026-10-12T18:00:00Z",
  "status": "scheduled"
}
```

14. VAPT Security Requirements

All endpoints, interfaces, and stored data must comply with the following security controls. A quarterly VAPT (Vulnerability Assessment & Penetration Testing) must be conducted by a CERT-IN empanelled vendor.

14.1 Authentication & Session Management

Control	Requirement	Implementation
2FA Enforcement	All admin panel logins require TOTP (Google Authenticator)	speakeasy / pyotp
JWT Tokens	Short-lived access tokens (15 min) + refresh token (7 days)	HS256 signed, stored in httpOnly cookie
Password Policy	Min 12 chars, uppercase, number, special char	zxcvbn strength library
Brute Force Protection	Lock account after 5 failed logins for 30 min	Redis counter per IP + user
Session Invalidation	All active sessions invalidated on password change	Token blacklist in Redis
OTP Security	OTPs expire in 5 min, single-use, 6-digit minimum	Backend enforced
IP Whitelisting	Admin panel optionally restrict to corporate IPs	Nginx / WAF rules

14.2 API Security

Control	Requirement	Implementation
HTTPS Everywhere	All endpoints TLS 1.2+, HSTS enabled	SSL cert + Nginx config
Rate Limiting	/auth endpoints: 10 req/min; /api: 200 req/min per user	Flask-Limiter + Redis
Input Validation	All inputs validated server-side (type, length, regex, injection)	marshmallow / pydantic
SQL Injection Prevention	ORM only (SQLAlchemy), no raw SQL strings	Parameterized queries
XSS Prevention	Output encoding, Content-Security-Policy header	Flask + React escaping
CSRF Protection	SameSite=Strict cookie, CSRF token on state-changing endpoints	Flask-WTF / custom

Control	Requirement	Implementation
IDOR Prevention	Always validate resource ownership in every query	Owner check in every handler
Sensitive Data Masking	Aadhaar, PAN, bank account masked in all API responses	Mask last digits
API Versioning	Sunset old API versions, enforce version in path	/api/v1/, /api/v2/

14.3 Data Security

Control	Requirement
PII Encryption at Rest	Aadhaar, PAN, bank account numbers encrypted with AES-256 in DB
Document Storage	KYC docs in private S3 bucket; presigned URLs with 10-min expiry
Database Hardening	Separate DB user per service, minimal privileges, no remote root access
Backup Encryption	Daily encrypted backups to offsite S3, retained 30 days
Log Sanitization	No PII in application logs; mask phone/email in logs
Data Retention Policy	Inactive user data purged after 3 years; audit logs retained 7 years

14.4 OWASP Top-10 Checklist

OWASP Risk	Mitigation
A01 Broken Access Control	RBAC enforced at API layer; ownership checks on every resource
A02 Cryptographic Failures	AES-256 at rest, TLS 1.2+ in transit, no MD5/SHA1
A03 Injection	ORM only, parameterized queries, input validation
A04 Insecure Design	Threat modelling, security review in design phase
A05 Security Misconfiguration	Hardened Docker images, no debug mode in prod, secrets in Vault
A06 Vulnerable Components	Automated dependency scanning (Snyk / Dependabot), monthly patches
A07 Auth Failures	2FA, brute-force protection, secure session management
A08 Software Integrity	Signed container images, CI/CD pipeline integrity checks
A09 Logging Failures	Centralised logging (ELK), alerting on anomalous patterns
A10 SSRF	Block internal IPs in outbound requests, URL allowlist

14.5 Audit Logging Requirements

Every action in the admin panel must be logged immutably with:

- WHO: admin_user_id, role, IP address, device fingerprint
- WHAT: action type (create/update/delete/approve/export), resource type, resource ID
- WHEN: UTC timestamp (microsecond precision)
- RESULT: success / failure + error code
- BEFORE/AFTER: snapshot of changed fields (for updates)

Audit Log Schema

```
{
  "log_id": "log_uuid",
  "admin_user_id": "adm_xyz",
  "role": "manager",
  "ip_address": "203.xxx.xxx.xxx",
  "user_agent": "Mozilla/5.0...",
  "action": "driver.approve",
  "resource_type": "driver",
  "resource_id": "drv_abc",
  "before": {"status": "under_review"},
  "after": {"status": "active"},
  "result": "success",
  "timestamp": "2026-03-01T10:00:00.123456Z"
}
```

15. Future Roadmap Features

15.1 Phase 2 (6-12 Months)

Feature	Description	Priority
AI-based Driver Matching	ML model assigns best driver based on rating, proximity, acceptance history	High
EV Fleet Management	Track battery %, charging stations, range anxiety alerts	High
Subscription Plans (Users)	Monthly pass: unlimited rides within city for fixed fee	High
Rental & Outstation Rides	Multi-hour/day bookings, inter-city travel	High
Driver Incentive Gamification	Badges, leaderboard, streak bonuses to improve retention	Medium
Chat Support (In-App)	User ↔ Driver real-time chat during ride	Medium
Multi-language Support	Telugu, Hindi, Tamil for driver & user apps	High
Auto BGV Integration	API-based background verification (Aadhaar e-KYC, DigiLocker)	High
Corporate Accounts	Business travel booking, monthly billing, travel policies	Medium

15.2 Phase 3 (12-24 Months)

Feature	Description	Priority
RailCab Marketplace	Vendors compete on price for inter-city routes	Medium
Insurance Integration	Per-ride micro-insurance for passengers	Medium
ONDC Integration	Mobility layer on Open Network for Digital Commerce	High
Ride Pooling	Shared rides along similar routes (cost splitting)	Medium
Driver Finance Products	Vehicle loans, EMI tracking, income advances via NBFC partners	Low
Predictive Surge	ML model predicts surge 30 min ahead based on historical patterns	Medium
Fuel Management	Vendor fuel card integration, expense tracking	Low
WhatsApp Booking	Book rides via WhatsApp chatbot	Low

Feature	Description	Priority
Safety SOS Integration	Emergency button triggers police & emergency contacts	High
Carbon Footprint Tracker	Show CO2 saved for EV rides	Low

16. API Summary Index

Admin Panel APIs

Method	Endpoint	Module	Auth Required
GET	/admin/dashboard/summary	Super Admin	Admin JWT + 2FA
GET/PATCH	/admin/config	Super Admin	Super Admin only
POST	/admin/roles	RBAC	Super Admin
POST	/admin/users	RBAC	Super Admin
GET	/admin/drivers/pending-approval	Manager	Manager/Admin
POST	/admin/drivers/:id/approve	Manager	Manager/Admin
POST	/admin/drivers/:id/reject	Manager	Manager/Admin
GET	/admin/rides	Manager	Manager/Admin/Accountant/Support
GET	/admin/earnings/driver/:id	Accountant	Accountant/Admin
POST	/admin/payouts/initiate	Accountant	Accountant/Admin
GET	/admin/gst-report	Accountant	Accountant/Admin
POST	/admin/support/tickets	Support	Support/Manager/Admin
POST	/admin/support/tickets/:id/resolve	Support	Support/Manager/Admin
GET	/admin/analytics/rides	Analytics	Manager/Admin/Accountant
GET/PATCH	/admin/fare-config	Fare Engine	Manager/Admin
POST	/admin/notifications/broadcast	Notifications	Manager/Admin
GET	/admin/vehicles	Fleet	Manager/Admin
GET	/admin/audit-logs	Security	Super Admin/Auditor

Vendor APIs

Method	Endpoint	Description
POST	/vendor/register	Vendor self-registration
POST	/vendor/:id/documents	Upload business documents
GET	/vendor/:id/dashboard	Vendor KPI dashboard
POST	/vendor/:id/vehicles	Add vehicle to fleet
GET	/vendor/:id/vehicles	List fleet vehicles

Method	Endpoint	Description
GET	/vendor/:id/earnings	Earnings breakdown
GET	/vendor/:id/payouts	Payout history
POST	/vendor/:id/drivers	Initiate driver onboarding under vendor

Driver Onboarding APIs (Enhanced)

Method	Endpoint	Description
POST	/driver/send-otp	Send OTP to driver phone
POST	/driver/verify-otp	Verify OTP and create account
POST	/driver/register	Submit personal + vehicle info
POST	/driver/:id/documents	Upload all KYC documents
GET	/driver/:id/onboarding-status	Check current onboarding step & status
POST	/driver/:id/safety-deposit	Record safety deposit payment
GET	/driver/:id/profile	Full driver profile
PATCH	/driver/:id/profile	Update driver profile

Core Booking APIs (Existing — Enhanced)

Method	Endpoint	Enhancement Needed
POST	/booking/calculate-fare	Add surge, night charge, weather factor in response
POST	/booking/request-ride	Add fare breakdown, estimated arrival time
POST	/booking/accept-ride	Return driver real-time ETA
POST	/booking/otp-verification	Return route polyline for navigation
POST	/booking/driver/end-ride	Trigger payment, update earnings, send invoice
POST	/booking/user/cancel-ride	Apply cancellation charge if applicable
POST	/booking/driver/cancel-ride	Apply driver penalty, requeue to next driver

END OF DOCUMENT — RailCab Admin Panel Requirements v1.0

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